

ANTI-MONEY LAUNDERING (AML) TRANSACTION MONITORING THROUGH INTELLIGENT MACHINE LEARNING ARCHITECTURE

Author: UPADRASTHA VIDHATH

Email: upadrastavidhath@gmail.com | GitHub: @UPADRASTHAVIDHATH

ABSTRACT

Keywords: Anti-Money Laundering (AML), Financial Crime Detection, Machine Learning, Random Forest, Structuring/Sn

PROJECT METHODOLOGY & KEY HIGHLIGHTS

- Behavioral Feature Engineering: Extracted transaction velocity (24h/7d), balance discrepancies, and structuring indicators.
- Imbalanced Learning Optimization: Addressed extreme class imbalance (2.3% positive rate) using stratified sampling and cost-sensitive learning.
- Algorithm Benchmarking: Evaluated Logistic Regression, Decision Trees, Random Forest, and Gradient Boosting.
- Explainable AI: Generated feature importance rankings aligning with FinCEN and FATF compliance red flags.
- Production Inference & SAR: End-to-end module for real-time risk tiering and automated Suspicious Activity Report generation.